

Oct 4, 2026 · @N

## Question and recommendation

A sponsor should take FID on a 100 MW onshore wind farm in Finland (COD 2028) only with a corporate PPA of at least 15 years at about 70 EUR/MWh or more. At the base-case PPA price of 50 EUR/MWh the project has a nominal NPV of –46.5 EUR m at a 6 % WACC; without a PPA it is not bankable.

The question behind this memo is what would have to be true for the project to reach FID. Three answers follow from the model:

- **Fully merchant does not finance under base assumptions.** Without a PPA, lenders size debt to only 17 % of CAPEX (47 % with a PPA). The best merchant scenario reaches 2.1 % equity IRR.
- **The PPA price decides.** The break-even PPA price is 70.35 EUR/MWh (NPV = 0) to 73.94 EUR/MWh (10 % equity IRR). Public PPA references sit at 29–50 EUR/MWh, and no realistic cost lever closes the gap on its own.
- **The battery is a separate bet.** A co-located 25 MW / 50 MWh battery is evaluated standalone. It needs 78 kEUR/MW/yr of gross revenue vs 89 today and does not change the wind economics.

This memo explains the market view, method, results and limits behind these numbers. The accompanying deck carries the exhibits; the Excel model (v1.8) and the Python data pipeline carry the calculations.

## Market view

Finnish wind earns about 60 % of the average day-ahead price, and this cannibalisation, not the price level, is the core market risk.

**Price level.** Day-ahead prices are skewed: in 2025 the mean was 40.5 EUR/MWh but the median only 16.7 EUR/MWh. The model's base path uses EEX futures to 2030 (Cal-28 56.02 and Cal-30 52.68 EUR/MWh nominal, trading day 1 Oct 2026; 53.84 and 48.67 in real 2026 EUR) and the only public long-term forecast for Finland from 2040 (Lampela 2023, MSc thesis, Tampere University). Over 2027–36 this path averages 48.2 EUR/MWh real, 6–12 EUR/MWh below the KYOS baseload outlook of 60.0 EUR/MWh, so the base case is conservative.

**Cannibalisation.** The wind capture rate (wind-weighted over time-weighted day-ahead price, fleet profile, ENTSO-E data) fell as wind capacity in Fingrid's forecast data grew by about 60 %:

| Period            | Capture rate | No output at negative prices | Wind energy in negative-price hours |
|-------------------|--------------|------------------------------|-------------------------------------|
| 2023              | 71.6 %       | 80.6 %                       | 8.2 %                               |
| 2024              | 65.5 %       | 74.5 %                       | 11.5 %                              |
| 2025              | 55.1 %       | 59.1 %                       | 6.5 %                               |
| Oct 2025–Sep 2026 | 61.8 %       | 63.0 %                       | 1.8 %                               |

The mechanism is visible hour by hour: over the last 12 months the calmest fifth of hours averaged 102 EUR/MWh, the windiest fifth 17 EUR/MWh. Output is also winter-weighted (December to March each 9–11 % of annual energy, July 4 %), which matters for a data-centre offtaker with a flat load.

**Cross-check.** The KYOS outlook for 2027–36 implies a Finnish wind capture rate of  $36.1 / 60.0 = 60.2$  %, matching the model base of 60 % and the lowest of all zones in the KYOS table (SE3 85 %, DK1 88 %, DE 83 %). The model uses the variant without output at negative prices, consistent with its 5 % curtailment, to avoid double counting.

Sources: own analysis of [ENTSO-E Transparency Platform](#) data (accessed 1 Oct 2026); Fingrid open data, dataset 268; EEX market data; Lampela, S. (2023); KYOS PPA Insights No. 19 (Jun 2026).

## Model and method

The Excel model is a project-finance model of a single wind SPV, built close to the two recent Finnish FIDs by OX2, both fully contracted with corporate offtakers.

**Inputs.** Prices and costs are entered in real 2026 EUR and indexed at 2 % (ECB medium-term target); WACC and debt rate are nominal. Every input carries a source and one of four status flags: verified, source-based, own assumption, to verify. Of 43 active wind inputs, 3 are verified, 24 source-based and 16 own assumptions.

| Input                   | Base case              | Status                        |
|-------------------------|------------------------|-------------------------------|
| Capacity, COD, lifetime | 100 MW, 2028, 25 years | own assumption / source-based |

| Input               | Base case                                                  | Status                                |
|---------------------|------------------------------------------------------------|---------------------------------------|
| Net capacity factor | 35 % (P50)                                                 | source-based                          |
| Curtailement        | 5 %                                                        | source-based                          |
| CAPEX               | 1,500 EUR/kW                                               | source-based                          |
| OPEX incl. lease    | 44 EUR/kW/yr                                               | source-based                          |
| PPA                 | 100 %, 15 years, 50 EUR/MWh fixed nominal                  | source-based / own assumption (price) |
| Debt                | 5.75 %, 15 years, minimum DSCR 1.30x, gearing cap 75 %     | source-based / own assumption (DSCR)  |
| WACC                | 6 % with PPA, 7 % merchant (CAPM build-up $\approx$ 5.9 %) | own assumption                        |
| Tax                 | 20 %, depreciation 23.5 % declining balance                | verified / source-based               |

**Revenue.** PPA volume earns the fixed nominal price for 15 years; after expiry the plant sells at the selected market price path times the capture rate. Market price level and capture rate are two independent Low/Base/High switches.

**Debt.** Debt is sculpted to a target DSCR on cash flow available for debt service after unlevered tax, which avoids circularity, and capped at 75 % gearing. DSCR and WACC are set per scenario: 1.30x and 6 % with a full PPA, 1.60x and 7 % merchant.

**Break-even.** The break-even PPA price is solved by iteration on the PPA price with full recalculation, including debt re-sizing, because tax losses and the gearing cap make NPV non-linear in price. A linear approximation would overstate it by about 1.4 EUR/MWh.

## Results

At 50 EUR/MWh the project destroys value; it needs about 70–74 EUR/MWh, and the PPA price is the only single lever that closes that gap.

**Base case.** NPV –46.5 EUR m, project IRR 1.6 %, equity IRR –0.3 % (nominal), LCOE 47.9 EUR/MWh (real 2026), gearing 47 %. The LCOE is below the PPA price, yet value is negative: a fixed nominal 50 EUR/MWh is worth only about 43 EUR/MWh in real 2026 terms over 2028–2042 (own calculation, 6 % discount rate), below the real LCOE, and after 2043 the plant sells at about 60 % capture.

**PPA vs merchant.** Across nine price × capture scenarios, equity IRR ranges from –5.4 % to 2.6 % with a full PPA and from –12.1 % to 2.1 % merchant. Under the PPA the range is narrow because the scenarios only act after 2043. Without a PPA, debt capacity is the binding constraint: a 1.60x DSCR on merchant cash flow supports 17 % gearing.

**Break-even.** NPV = 0 at 70.35 EUR/MWh, where project IRR equals the 6 % WACC and equity IRR is 7.5 %, close to the CAPM cost of equity of 7.0 %. A 10 % equity IRR needs 73.94 EUR/MWh, where the 75 % gearing cap binds. The gap to the base case is 20–24 EUR/MWh. Public references are lower: KYOS P25 PPA estimate 28.9, Pexapark Euro Composite 45 (Jul 2026, all technologies), LevelTen P25 wind offers for Finland and Portugal 49.95 EUR/MWh (Q1 2026; Finnish wind steady in Q2 2026). These references are directional, not like-for-like: product, tenor, shape, country and offtaker credit differ.

**Sensitivities.** Single levers around the base case, project NPV in EUR m:

| Lever (unfavourable / favourable)     | NPV low | NPV high | Spread |
|---------------------------------------|---------|----------|--------|
| PPA price 25 / 80 EUR/MWh             | –111.3  | +21.3    | 132.6  |
| CAPEX 1,650 / 1,250 EUR/kW            | –60.0   | –24.3    | 35.7   |
| OPEX ±30 %                            | –64.5   | –29.4    | 35.1   |
| Capacity factor 32 / 38 %             | –59.2   | –34.1    | 25.1   |
| WACC 8 / 5 %                          | –58.9   | –38.6    | 20.3   |
| PPA volume 65 % (DSCR 1.40x)          | –61.7   | –        | 15.3   |
| Curtailment 10 / 2 %                  | –54.2   | –41.9    | 12.4   |
| Debt rate 6.75 / 4.75 % (WACC linked) | –49.1   | –43.6    | 5.5    |

Only with all seven cost and financing levers favourable at once does NPV turn positive (+20.4 EUR m, 11.0 % equity IRR), which is unrealistic (own judgement). Longer tenor and lower tax hardly matter: a 20-year PPA adds 5.0 EUR m, an 18 % tax rate 0.2 EUR m.

**Path to FID.** The required PPA price falls by about 3.5–3.8 EUR/MWh per 100 EUR/kW of CAPEX saved and by about 8 EUR/MWh with a better site (capacity factor 38.7 % instead of 35 %). With OX2-type parameters (about 1,480 EUR/kW, 38.7 %) it is about 62 EUR/MWh; at 1,250 EUR/kW and 38.7 % about 54. Cost and site narrow the gap but do not bring the required price to the level of public offers. Scale helps less through CAPEX per MW (OX2: 277 MW at 1.44, 472 MW at 1.48 EUR m/MW) than through O&M and access to hyperscaler deals; there is no evidence that it raises the PPA price.

## Co-located battery

A co-located 25 MW / 50 MWh battery pays its 7 % hurdle only if revenues stay near today's level; it is a separate merchant investment, not a fix for the wind case.

**Set-up.** The battery is evaluated in its own sheet: grid-charged, merchant, 100 % equity, after tax, 15 years, CAPEX 287 EUR/kWh (Modo Energy 2026 capex survey, global median for 2-hour projects). It is kept outside the wind SPV so that its merchant cash flows do not distort the wind debt sizing.

| Case                        | Gross revenue (kEUR/MW/yr) | NPV at 7 % (EUR m) | IRR (nominal) |
|-----------------------------|----------------------------|--------------------|---------------|
| Low                         | 45                         | -6.5               | -1.5 %        |
| Base                        | 65                         | -2.5               | 4.1 %         |
| High = June 2026 level      | 89                         | +2.1               | 9.4 %         |
| Break-even                  | 77.8                       | 0.0                | 7.0 %         |
| Base with 20 % CAPEX saving | 65                         | +0.4               | 7.5 %         |

**Market.** The Clean Horizon 2-hour index for Finland stood at 89 kEUR/MW/yr in June 2026 and fell again in July as FCR and mFRR capacity prices dropped by about 30 %. Installed capacity is about 1.5 GW, and Fingrid connection agreements point to about 4 GW. A decline of more than about 13 % from the June level makes the battery uneconomic, so the base case assumes lifetime revenues 25 % below today.

**Co-location.** The battery does not change any wind cash flow. Charging from own wind carries the market price as opportunity cost, and Fingrid storage energy fees are only about 1 EUR/MWh, so behind-the-meter charging adds little. The real synergy is shared infrastructure: each 10 % CAPEX saving from a shared grid connection lowers the break-even revenue by about 7.3 kEUR/MW/yr. The size of that saving is an own assumption and not verified.

## Validation and AI use

The results were checked against independent sources and hand calculations in 14 logged checks; all 27 model integrity checks pass.

| Check                                            | Result                                                                                |
|--------------------------------------------------|---------------------------------------------------------------------------------------|
| Capture rate vs KYOS fundamental model           | 60.2 % vs 60 % base                                                                   |
| LCOE hand calculation (real annuity)             | 47.8 vs 47.9 EUR/MWh                                                                  |
| LCOE bridge to IRENA Finland 2025 (38.9 EUR/MWh) | Gap explained by CAPEX, capacity factor and WACC                                      |
| Break-even plausibility                          | Project IRR = WACC at NPV = 0                                                         |
| Wind data: Fingrid vs ENTSO-E                    | Within about 1 % from 2024; frozen Fingrid values in Jun–Sep 2023 found, ENTSO-E used |
| Recalculation and IRR cross-check                | Identical results in 18 scenario runs                                                 |
| Tornado and break-even grid reproduced           | Full recalculation of the current model version                                       |

**AI use.** The author used Claude (Anthropic) for four tasks: an audit of the original model template (four formula errors found and fixed), source search, drafting the Python data pipeline, and scripted scenario and break-even runs. Every input value was entered by the author after reading the source, with source text and status in the model. AI output was checked by hand calculation, independent recalculation and spot checks of the code against raw data. The model's Sources & Checks sheet logs each check and each AI use.

## Limitations and next steps

The conclusion is robust to the main simplifications, because the gap of 20–24 EUR/MWh is large relative to any single assumption.

### Limitations.

- Long-term prices rest on one MSc thesis using a neural network; its price basis (real or nominal) is not stated.
- Price level and capture rate are independent switches, although low prices and low capture rates tend to coincide.
- Financing is simplified: P50 only, a single DSCR, no interest during construction, fees or DSRA.
- The capture rate uses the fleet profile as a proxy for one park and ignores balancing costs.
- The OX2 break-even of about 62 EUR/MWh is an own derivation; actual PPA prices are not published.

- Battery O&M (1.5 % of CAPEX) and optimiser fee (8 % of revenue) are own assumptions to verify.

**Open regulatory points.** Government proposal HE 151/2026 (17 Sep 2026) cuts corporate tax to 18 % from tax year 2027 and extends loss carry-forward from 10 to 25 years for losses from tax year 2026; it is not yet law. The model keeps the current 20 % rate and carries losses forward without expiry, consistent with the proposal. Own sensitivities: 18 % adds about 0.2 EUR m NPV; a current-law 10-year expiry costs about 1.2 EUR m. The EUR 3m safe harbour for interest paid to unrelated lenders is not applied (conservative; project NPV unaffected).

### **Next steps.**

1. Lender case vs sponsor case: P90 generation, DSCR on P90, LLCR.
2. Construction financing: drawdown schedule, interest during construction, fees and a DSRA.
3. Tax update to the proposed 2027 regime, including the EUR 3m safe harbour and carry-forward of disallowed interest.
4. Hybrid offer: wind plus storage, shaped to a data-centre load. LevelTen's new Hybrid Index shows an average P25 of 71.58 EUR/MWh in Q2 2026, inside the break-even range, though mostly from solar plus storage in Spain and Germany.
5. Linked price and capture-rate scenarios instead of independent switches.

**Sources (selected).** [LevelTen European PPA Price Index Q2 2026](#) · [Finnish Government budget proposal 2027](#) · [Clean Horizon index June 2026 via ESS News](#) · [Clean Horizon index July 2026 via ESS News](#) · KYOS PPA Insights No. 19 (Jun 2026) · Lampela, S. (2023), Tampere University · OX2 press releases (1 Apr and 16 Dec 2025). The full source list is in the model's Inputs and Sources & Checks sheets.